

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAI001
TITLE OF THE COURSE : ENTERPRISE IT SYSTEMS AND APPLICATIONS
L: T: P: C : 3: 0: 2: 4

Overview

For the modern businesses operating in a competitive economy, the key requirements to succeed are operational efficiency, customer focus, agility, employee retention, financial prudence etc. Creative adoption of information technology tools and solutions to digitalize the business functions and processes is an essential ingredient to achieve the above elements. For managers / entrepreneurs a deep knowledge of information technology systems and solutions available for each Business Function, in the form of Enterprise Applications is essential. The tools available, technology trends, the key features, how to use these tools effectively, how to automate the business processes using these tools, how to use the right tools for the organization, etc. are extremely important to be effective in their role. This course intends to enhance the skill of managers/entrepreneurs by providing a holistic understanding of Enterprise Resource Planning and related functional systems for smooth functioning of an Enterprise.

Course Objectives

1. To provide deep understanding of Enterprise Resource Planning and related Functional Systems and Processes in a digital economy
2. To choose and use the right solutions to meet the IT specific Functional needs of a modern Business Enterprise.
3. To examine the importance of the applications of ERP in terms of Sales, Marketing, Finance & Accounting, Customer Relationship Management, Human Resource Management and Supply Chain Management.
4. To discuss all of the above in the context of readiness of an Enterprise for implementation of an ERP
5. To select an ERP and steps involved in the process of implementation.

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of enterprise applications and their critical role in business enterprises in the Enterprise IT Systems and Applications course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of enterprise technology solutions for selecting and using appropriate systems in the Enterprise IT Systems and Applications course.
3. Demonstrate the ability of analyzing real-time problems using skill acquired by concepts, theories, models and principles of ERP functional modules such as Sales and Marketing,

Finance and Accounting, CRM, SCM, and HR in the Enterprise IT Systems and Applications course.

4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of enterprise readiness and ERP implementation strategies in the Enterprise IT Systems and Applications course.
5. Exhibit the ability of developing or redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of ERP implementation methodologies in the Enterprise IT Systems and Applications course.

Approach to Learning	● Lectures ● Readings ● Active student participation and class room exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	Business Functions and Business Processes, Introduction to ERPs Functional Architecture of an Enterprise, Business Functions and Business Processes, Business Process Reengineering (BPR). ERP, Introduction – integrated management information; Scope and Benefits; Evolution; ERP implementation; Post implementation options.	8
Unit II	The Development of Enterprise Resource Planning Systems Evolution of Information Systems, Bespoke and Commercial Off the Shelf (COTS) Systems, ERPs. Key EPR Systems and their evolution, SAP R/3, Peoplesoft, Oracle. Cloud-based and SaaS ERPs. ERPs for small and medium Enterprises Choosing and ERP and Consulting Partner Relevant case studies of implementation of SAP R/3, Peoplesoft, Oracle.	8

Unit III	Marketing Information Systems and Sales Order Process, Customer Relationship Management Unintegrated Marketing and Sales processes – Problems with Fitter Snackers ‘Sales Processes. Sales & Distribution Processes in an ERP – Presales Activities, Sales Order Processing, Inventory Sourcing, Delivery, Billing and Payment. A Standard Order in SAP, ERP, Customer Relationship Management: Core CRM activities, SAP ‘s CRM Software and Benefits of CRM systems	8
Unit IV	Production Overview through a case study – Fitter’s Production & Manufacturing Processes and issues. Production Planning Process and SAP ‘s approach – Sales Forecasting, Operations Planning, Demand Management, Material Requirements Planning (MRP), Bill of Materials, Lead Times & Lot Sizing. Material Requirements Planning in SAP R/3	8
Unit V	Accounting Systems and Human Resource Management Systems Accounting activities and using and ERP for Accounting Information System. Operational Decision Making through Accounting Information Systems, Credit Management, Product Profitability Analysis, Inventory Cost Accounting, Management Reporting using ERPs, Case Studies such as Collapse of Enron Human Resource Processes within ERPs, Recruitment Management, Time Management, Payroll Processing, Travel Management Training and Development, Coordination, Management of objectives and employee performance, administration of employee leaves. Relevant case studies.	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	3	3	3	3
CO 2	3	2	3	2	3
CO 3	2	3	2	2	2
CO 4	2	3	3	1	2
CO 5	2	3	2	2	2

Action Based Component

- Video on how to implement a new ERP system
- Mini Project on Study on ERP Implementation at a University
- Do research on various COTS system
- Leave System – Requirements

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10		X	X			
7	Semester End Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Concepts in Enterprise Resource Planning (Fourth Edition), Ellen F. Monk, Bret J. Wagner, Cengage Learning 's
2. Accounting Information Systems (11th Edition), Iric J. Gelinas, Richard B. Dull, Patrick Wheeler, Mary Callahan Hill, Cengage

Reference books

1. Enterprise Resource Planning by Alexis Leon, Tata McGraw-Hill
2. Marketing Information System by C Gotlagunta, Neha Publishers

3. Human resource Information Systems by Michael J Kavanagh, Mohan Thite, Richard D Johnson, SAGE SOUTH ASIA EDITION

Readings & Case Analysis

1. Harvard Business Review- Surviving SAP Implementation in a Hospital By: Juanita Cajiao; Enrique Ramirez
2. The Diet Centre: The SAP ERP Decision
3. HBR Article:: Libor's Real Scandal: Bad Data By: Thomas C. Redman
4. Putting the Enterprise Into the Enterprise System, By: Thomas H. Davenport

Suggested Courses

1. Digital Marketing course by Google
2. Accounting information systems
3. Human resource Information System by Udemy

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAI002
TITLE OF THE COURSE : BUSINESS TECHNOLOGIES
L: T: P: C : 3: 0: 2: 4

Overview

Successful Business Models in the current times are based on the IT strategy of an organization. Some of the most successful new age enterprises have their Business Strategy entirely driven by Information Technology. New-Age Business Technologies have rapidly transformed the business landscape and competition. Businesses, which do not adapt to rapidly transformation in technology, lag behind or cease to exist. Therefore, it is imperative that new age Business Management professionals have a conceptual understanding and know-how of some of the new age technologies which are transforming the Business Landscape in the current.

Course Objectives

1. To learn about the contemporary technologies those are transforming the business landscape.
2. To learn the business applications of contemporary technologies.
3. To get a general understanding of tools and platforms used in the implementation of new age technologies
4. To understand the associated risks and mitigation in the implementation of new technologies
5. To understand the future applications of the new technologies

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of contemporary business technologies used for problem-solving in the Business Technologies course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of business applications for effective decision-making in the Business Technologies course.
3. Demonstrate the ability of analyzing real-time problems using skill acquired by concepts, theories, models and principles of emerging technologies transforming businesses in the Business Technologies course.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of technology risk

assessment and mitigation strategies in the Business Technologies course.

5. Exhibit the ability of developing or redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of technology integration and solution design in the Business Technologies course.

Approach to Learning	• Lectures • Readings • Active student participation and class room exercises • Case Analysis collaboratively with students involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	Introduction to Business technologies Introduction to following technologies: Cloud computing, Block chain, Robotic Process Automation (RPA), Internet of Things (IoT), Cyber Security, Virtual and Augmented Reality (AR/ VR), Artificial Intelligence and Machine Learning (AI/ ML), Evolution of technologies, the need in the industry for these technologies, future landscape and core theoretical concepts related to each technology. Augmented Reality and Virtual Reality, Concept of AR and VR, the difference between the two, underlying concepts, business application using a Case Study.	8

Unit II	Cloud Computing, Cyber Security Introduction to Cloud Computing and the need Cloud Computing Concepts: Containers, Micro services, Big data and analytics, Cloud Security, Cloud Migration, Public and Private Cloud. Cloud computing on Amazon Web Services (AWS) Cloud computing on Azure (Microsoft Cloud). Google Cloud Platform Cyber security: Introduction to Information Security, Protecting Computer Systems and Network, Identification of threats, attacks, intrusion etc. Malware, phishing, denial-of-service, password attacks Ransom ware and its prevention Tools and technology for Cyber security, Introduction to Cryptography and its application in securing data.	8
Unit III	Block chain, Internet of Things (IoT) What is Block chain, difference between Database and Block chain, Concept of Distributed Ledger, Architecture of a Block chain, Cryptography and Hashing, Intermediaries, Block chain Transaction Mechanism, Crypto currencies – an application of Block chain, Internet of Things (IoT), Core concepts, benefits, future of IoT, Business Application.	8
Unit IV	Robotic process Automation Robotic Process Automation, underlying concepts, popular tools & platforms (Ui path), benefits of RPA, issues in implementation, business application using a Case Study. RPA assignment using Ui Path, Intelligent Automation: RPA combined with AI (NLP and ML)	8
Unit V	Machine Learning & Artificial Intelligence Artificial Intelligence and Machine Learning, Core concepts, benefits, future of AI, Business Application, Responsible AI, Ethics, and Bias in Business Applications	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	2	2	3	2	1
CO 2	2	2	2	2	1
CO 3	3	2	2	2	1

CO 4	2	2	2	2	1
CO 5	1	1	2	2	1

Action Based Component

- Do research on Cloud Computing Service Providers
- Group assignments on application of Block chain
- Mini Project on RPA
- Video on Business application of AI

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10		X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20		X				
6	CBT	10		X	X			
7	Semester End Examination	40		X	X	X	X	

Recommended Resources

Textbook

1. Principles of Information Security, Michael E Whitman, Herbert J Mattord, Principles of information Security, Cengage Learning
2. An introduction to Internet of Things: Connecting Devices, Edge Gateway, and Cloud with Applications, Cengage Learning

Reference books

1. The Robotic Process Automation Handbook, Tom Taulli, Apress
2. The management of technological innovation, Mark Dodgson, David Gann and Ammon Salter, Oxford
3. Management Technology, Hans J Thamhain, Wiley

Readings & Case Analysis

1. Harvard Business Review
2. MIT Sloan Management Review
3. ISTD. Indian society for Training & Development

Suggested Courses

1. Introduction to artificial intelligence by Coursera

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAI003
TITLE OF THE COURSE : PROGRAM AND PROJECT MANAGEMENT,
CUSTOMER SERVICES MANAGEMENT
L: T: P: C : 3: 0: 2: 4

Overview

While it is well understood that information technology systems are the backbone of modern enterprises, how to effectively deploy these IT solutions is a generally overlooked area. The key elements of an exercise to deploy these IT solutions are centered around aligning the solution with the business needs, effective project management, synchronizing Project management activity with the complexities of software development, Process redesign, etc. This course prepares the managers/entrepreneurs to be the key players in deployment of IT solutions in businesses.

Course Objectives

1. To develop an understanding of the principles and techniques used in information systems development
2. To know the tools and techniques of Project management with special focus on Software services
3. To understand the tools and techniques used in customer services management
4. To introduce the students to the different methods and techniques used for Project Management
5. To provide an understanding of trends and adoption of these tools to meet the unique challenges of modern enterprises and digital economy

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of software processes, development methodologies, and frameworks including requirement engineering, design, testing, and implementation in the Program and Project management, Customer Services Management course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of project management knowledge areas and process groups for effective IT project planning and execution in the Program and Project management, Customer Services Management course.
3. Demonstrate the ability of analyzing real-time problems using skill acquired by concepts, theories, models and principles of project integration and scope management to identify dependencies, constraints, and deliverables in the Program and Project management, Customer Services Management course.
4. Exhibit the ability of designing and redesigning existing and new models by using the

acquired skills from concepts, theories, models and principles of project quality, risk, and resource management to enhance IT project outcomes in the Program and Project management, Customer Services Management course.

5. Exhibit the ability of developing or redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of IT service management strategies and ITIL frameworks for service strategy, transition, operation, and continual improvement in the Program and Project management, Customer Services Management course.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	Introduction to Software Development Introduction to software, Software Processes, Software Framework, Software Development Life Cycle (SDLC) Models: Software development Models- Waterfall model, Evolutionary software development, Software prototyping, Incremental Model, Re-use oriented development	8
Unit II	Software Development Methodology and Estimation techniques Requirement Engineering, Documentation, Software Configuration and Change Management, Structured Analysis & Design, Software Architecture, Use Case User interface and User Experience design, Software Testing and Implementation, Software Cost Estimation, Function Point Analysis, COCOMO Model, Quality Management, Software Metrics	8
Unit III	Software Project Management Values, Manifesto, Methodologies, Scrum Framework, Team Management, Scaled Agile Frameworks (SAFe)	8

Unit IV	ITSM IT Services Management framework and concepts, ITIL 4 and Service Value System	8
Unit V	Cyber Law Software and Hardware Procurement, Installation, UI Testing, Training, Cyber Crime- Technical issues, Legal issues, Penalty under IT Act, Digital Signatures: Certificates, E-Governance in India; Contract in the Information Technology world, IT Act 2000, Jurisdiction under IT act 2000	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	2	3	3	1
CO 2	3	3	3	3	1
CO 3	3	2	3	3	1
CO 4	3	3	3	3	2
CO 5	3	2	2	3	2

Action Based Component

- Do research on software development models
- Group assignments on application of software metrics development
- Reflection based on guest talk on SCRUM Framework
- Mini Project on ITSM
- Video on software testing

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						

3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semest er Exam	20	X	X				
6	CBT	10		X	X			
7	Semester En d Examination	40	X	X	X	X	X	

Recommended Resources

Textbook

1. Software Project Management – Dr Sanjay Mohapatra, Cengage Learning
2. Software Management – A Practitioner's Approach by Roger S Pressman - McGraw-Hill

Reference books

1. Succeeding with agile software development using SCRUM by Mike Cohn- Pearson India
2. Project Management – Process, Technology and Practices- - Ganesh Vaidyanathan, Pearson
3. Cyber Law and IT Protection, Harish Chander- PHI

Readings & Case Analysis

1. Harvard Business Review – Case Study Broadband-X: Enterprise Resource Planning Implementation By: Mustafa Fatih Yegul
2. HBR Article: Why User Experience Always Has to Come First By: Michael Schrage

Suggested Courses

1. Project Management Software
2. Source Code management Software – Jenkins

SEMESTER/YEAR : III SEM / II YEAR
COURSE CODE : 25MBAI004
TITLE OF THE COURSE : STARTUP AND PRODUCT DEVELOPMENT:
(INNOVATION, IDEATION, PRODUCT MANAGEMENT,
INTELLECTUAL PROPERTY & BUSINESS
DEVELOPMENT)
L: T: P: C : 3: 0: 2: 4

Overview

This course is intended to enable students to understand how software products are conceptualized, designed, brought to market and scaled. The role of innovation in new product development is critical and there are informal and formal practices that are prevalent in the industry to approach innovation. Innovation in start-ups follows a different path vs. an established corporation. This course teaches the basics of ideation, innovation, product development and productization.

The course also teaches basics of intellectual property protection: legal means available and legal means compared with other more viable approaches.

Course Objectives

1. Learn the basics of new software product creation.
2. Gain basic understanding of techniques used by highly innovative software companies today.
3. Understand how to create a business model, minimum viable products and launching products in markets.
4. Understand the role of managing funds, securing funds in running a successful new product company.
5. Understand intellectual property rights related to software products.

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of market evolution and drivers of change in the Startup and Product Management course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of sectoral innovation and market dynamics in the Startup and Product Management course.
3. Demonstrate the ability of analyzing the real-time problems using skill acquired by concepts, theories, models and principles of innovation models and entrepreneurial practices in the Startup and Product Management course.
4. Exhibit the ability of designing and redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of intellectual property management and product scalability in the Startup and Product Management course.

5. Exhibit the ability of developing or redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of enterprise strategy, funding, and sustainability in the Startup and Product Management course

Approach to Learning	• Lectures • Readings • Active student participation and class room exercises • Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: • Mini projects, • Submission of assignments • Group assignments • Written Exam

Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	Market Dynamics and Opportunities for Innovation Introduction, Changing Market Dynamics in India, Alternative Business Models, Opportunities for Small New Businesses	8
Unit II	Introduction to Innovation & Fundamentals of New Business Creation Terminology and Key Definitions of Innovation, Successful Innovative Companies, Models of Innovation, Success Factors for Innovation, Game changing Innovations in the Software Industry, Design Thinking and Customer Centric Innovation	8
Unit III	Typical Software Product Lifecycle Ideation, Market Opportunity Identification, Business Model Canvas, MVP and MDE, Productization, Customer Acquisition and Scaling the Business, Product Decline and New Product Creation, Product–Market Fit and Metrics for Product Success	8
Unit IV	Intellectual property Protection Importance of Intellectual Property Protection, Property Law, Practical Approaches	8
Unit V	Building Your Company Essential Team Composition, Securing Funding and Managing Cash Flows, Mergers and Acquisitions, Surviving Downturns and Changing Market Dynamics	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	1	3	2	2	2
CO 2	2	2	2	2	1
CO 3	3	2	1	3	2
CO 4	1	1	3	2	2
CO 5	2	2	3	2	2

Action Based Component

- Do research on Changing Market Dynamics in India
- Group assignments on Innovations in Software Industry
- Mini Project on Creating a software Product
- Reflection based on guest talk on intellectual Property Law
- Video on building software Company

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10		X	X			

7	Semester En d Examination	40	X	X	X	X	X	
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Recommended Resources

Textbook

1. The Management of Technology and Innovation, Cengage eBook White, Bruton
2. Intellectual property Rights, Nithyananda K V, CENGAGE

Reference books

1. Grabbing Lightning: Building a Capability for Breakthrough Innovation, Gina O'Connor, 2008 WILEY Publications.
2. The Lean Startup: How Constant Innovation Creates Radically Successful Businesses, by Eric Ries, October 2011, Penguin UK.
3. Managing innovation: Integrating Technological, Market and Organizational Change, Joe Tedd, John R Beasant, Wiley Publications
4. Do the Work: Overcome Resistance and Get Out of Your Own Way, by Steven Pressfield and Seth Godin, March 2015, Black Irish Entertainment LLC, 2015.

Readings & Case Analysis

1. Harvard Business Review Case: CarMax: Driving What's Possible, By: Ryan Nelson; Ryan Wright
2. Harvard Business Review Article: Foundations for Growth: How to Identify and Build Disruptive New Businesses By: Clayton M. Christensen; Mark W. Johnson; Darrell K. Rigby

Suggested Courses

1. Software Project Management
2. Agile with Jira

SEMESTER/YEAR : IV SEM / II YEAR
COURSE CODE : 25MBAI005
TITLE OF THE COURSE : FUNDAMENTALS OF AI/ MACHINE LEARNING & RPA
L: T: P: C : 3: 0: 2: 4

Overview

This course is intended to enable students to understand the basics of Artificial Intelligence, Business Analytics and Robotic Process Automation. All the three areas are important in the industry today and deliver great business impact. The course covers fundamental techniques in AI and business applications for industry users. Similarly, Business Analytics fundamentals are covered from a user perspective with minimal theory and popular tools are used to provide a practical understanding.

Robotic Process Automation is delivering efficiencies for businesses today through the use of technology. The course covers the basics of the approach in RPA followed by usage of standard tools.

Course Objectives

1. Learn the fundamentals of Neural Networks, Machine Learning, Deep Learning and Natural Language Processing.
2. Understand and apply AI techniques to simple and illustrative industry problems.
3. Understand data collection, data warehousing, reporting, visualization
4. Deriving insights using BA tools such as Micro strategy.
5. Obtain a basic understanding of the importance and business impact of RPA and development automations using UIPath or BluePrism.

Course Outcomes

After completion of this course students will be able to:

1. Exhibit the fair understanding of concepts, theories, models and principles of Artificial Intelligence techniques and their business relevance in the Fundamentals of AI, Machine Learning & RPA course.
2. Demonstrate the ability of applying the skill of concepts, theories, models and principles of Artificial Intelligence for solving business problems in the Fundamentals of AI, Machine Learning & RPA course.
3. Demonstrate the ability of analyzing real-time problems using skill acquired by concepts, theories, models and principles of data warehousing and ETL tools for building business data repositories in the Fundamentals of AI, Machine Learning & RPA course.
4. Exhibit the ability of designing and redesigning existing and new models by using the

acquired skills from concepts, theories, models and principles of data visualization, reporting, and dashboard analytics in the Fundamentals of AI, Machine Learning & RPA course.

5. Exhibit the ability of developing or redesigning existing and new models by using the acquired skills from concepts, theories, models and principles of Robotic Process Automation for business process optimization in the Fundamentals of AI, Machine Learning & RPA course.

Approach to Learning	● Lectures ● Readings ● Active student participation and classroom exercises ● Case Analysis collaboratively with students' involvement
Assessment Strategy	Participants will be assessed on both conceptual understanding and business application of Finance practices by way of: ● Mini projects, ● Submission of assignments ● Group assignments ● Written Exam

Syllabus

<u>Units</u>	<u>Syllabus Details</u>	<u>Teaching Hours</u>
Unit I	AI for Business Users in Enterprises The meaning behind common AI terminology, including neural networks, machine learning, deep learning, and data science, building machine learning and data science projects, Using AI for business problems and defining the AI roadmap to help your company, collaborating with your IT team to get the best out of AI, Practical limitations and capabilities of AI	8
Unit II	Data Warehousing Basics Data Collection and Data Quality, Schema Design and Modelling Techniques, OLTP and ETL, Dimensions and Views, Partitioning and Parallelism, Change Data Capture for a Data Warehouse, Data Lake and Lakehouse Concepts	8
Unit III	Theoretical Understanding of data Analytics Probability and Statistics, Regression Analysis, Time Series Forecasting, Predictive Analytics, Prescriptive Analytics, Optimization and Simulation Modelling	8
Unit IV	Dash-boarding, Creating and deriving Insights Data Visualization using standard tools, creating meaningful Reports, Deriving Insights and Inferences from Reports, Data Storytelling and Decision-oriented Dashboards	8
Unit V	Robotic process Automation Introduction to RPA, Commonly Solved problems using RPA Hands-on exercises using RPA tools	8

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5
CO 1	3	1	1	1	2
CO 2	3	2	1	1	3
CO 3	1	2	1	3	3

CO 4	3	1	3	3	2
CO 5	2	2	2	2	2

Action Based Component

- Do research on AI capabilities
- Group assignments on data warehousing applications in select industries
- Video on data analytics
- Reflection based on guest talk on data visualization
- Mini Project on RPA

Course Assessment

#	Description of Assessment Method	Weightage %	Learning Outcomes Assessed					Submission day/week (assignments) or length (exam)
			1	2	3	4	5	
1	Class Participation	5						
2	Attendance	5						
3	Assignment 1	10	X	X				
4	Assignment 2	10			X	X		
5	Mid Semester Exam	20	X	X				
6	CBT	10		X	X			
7	Semester End Examination	40	X	X	X		X	

Recommended Resources

Textbook

1. Business Analytics: Data Analysis and Decision Making: Christian Albright and Wayne Wilson, Cengage Publications.

2. The Robotic process Automation Handbook, Tom Taulli, Apress

Reference books

1. Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, Pearson Publications.
2. Business Intelligence, Analytics, and data Science- Ramesh Shrada, DursenDelen, Efraim Turban, Pearson
3. Data warehousing fundamentals, Paulraj Ponniah, Wiley Publications

Readings & Case Analysis

1. Harvard Business Review Case Study: Flipkart vs. Amazon vs. Snapdeal: Winner- Take-All Battle in India by: Rainny Shuyan Xie; Vijay Sethi
2. HBR Article: Does Your Company Really Need a Chatbot?
By
: PV Kannan; Josh Bernoff

Suggested Courses

1. Data Analytics course
RPA Process automation